F. Take the L

Observations

For every point, there are two possible choices:

- **A** = right + up
- **B** = left + down

Consider two points p, q such that

$$x_p < x_q.$$

If

$$y_p < y_q,$$

the only forbidden combination is

$$p = A, \quad q = B.$$

If

$$y_p > y_q,$$

the points form an inversion, and they must have different types.

Therefore, after sorting by x, the problem can be represented using only the relative order of the y-coordinates.

Replace every y-coordinate by its rank to obtain a permutation p.

Inversion Observation

For $i < j$:

$$p_i > p_j$$

means that the two points form an inversion, so they must receive different types.

Thus, the inversion graph must be bipartite.

A decreasing subsequence of length 3,

$$p_i > p_j > p_k,$$

creates a triangle of inversions. A triangle cannot be colored using only two types, so such a subsequence is impossible.

Splitting into Blocks

The connected components of the inversion graph are contiguous.

While scanning the permutation, maintain:

$$pm = \max(p_0, p_1, \ldots, p_i).$$

Whenever

$$pm = i + 1,$$

the current component ends.

This is because the first $i+1$ positions then contain exactly the values

$$1, 2, \ldots, i+1,$$

so no element after this position can form an inversion with an element before it.

Number of Non-Trivial Blocks

A block of size 1 has no inversion.

A block of size greater than 1 contains an inversion, so it must contain both A and B.

If there are two non-trivial blocks, the first block is completely lower-left of the second block. Since the first contains an A and the second contains a B, a forbidden A-before-B pair is unavoidable.

Therefore:

$$\boxed{\text{There can be at most one non-trivial block.}}$$

If there are no non-trivial blocks, all points can have the same type, so the answer is `YES`.

The Unique Big Block

Suppose the only non-trivial block is

$$[L, R].$$

All singleton blocks before it are forced to be **B**.

All singleton blocks after it are forced to be **A**.

Therefore the overall structure is:

$$B\ B\ \cdots\ B \quad | \quad [L, R] \quad | \quad A\ A\ \cdots\ A.$$

Coloring the Big Block

Inside the big block, consider the prefix maxima.

A position i is a prefix maximum if

$$p_i > \max(p_L, \ldots, p_{i-1}).$$

All prefix maxima are increasing, so they do not form inversions with each other. They can therefore receive the same type.

Let these points form group X.

All remaining points form group Y.

Every point in Y has an earlier larger prefix maximum, so it must have the opposite type from X.

Therefore, there are only two possible colorings:

$$X = A, \quad Y = B$$

or

$$X = B, \quad Y = A.$$

Checking the Second Group

The points in Y must be increasing.

If two points of Y form an inversion, they would have the same type even though an inversion requires different types.

Therefore:

$$p_{Y_1} < p_{Y_2} < \cdots$$

must hold.

If this is not true, the answer is `NO`.

Final Condition

After fixing the types, the only remaining condition is that an A point cannot be both:

- to the left of a B point, and
- below that B point.

While scanning the points from left to right, maintain:

$$minA = \text{minimum } y \text{ among all previous A points.}$$

For a B point with coordinate y_i, if

$$minA < y_i,$$

then there is an earlier A point below it, so this assignment is invalid.

Try both possible colorings of X and Y. If either one passes the final check, the answer is `YES`.